Relic Gravitational Waves from Early-Universe Turbulence: Analytical Approach

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Current cosmological observations, including precise data from cosmic microwave background (CMB) anisotropies and large scale structure (LSS) surveys together with light element abundances, provide us with the universe picture from the moment of big bang nucleosynthesis (BBN) till today. However, the earlier moments remain hidden. On the other hand, testing these moments is an invaluable possibility to get information about fundamental physics acting on extreme energies [1–5].

The recent discovery of the stochastic gravitational wave background (SGWB) through pulsar timing arrays (PTAs) increased interest in GW astronomy. Even though it is unlikely that the signal is from the early universe, such a possibility is not excluded. The violent processes in the early universe just shortly after inflation, such as preheating/reheating, cosmological phase transitions and follow up primordial turbulence, and/or magnetic fields, e.g. any anisotropic, traceless and traverse source can produce gravitational waves detectable by future detectors [6–9]. The focus of this letter is to determine early-universe turbulence generated GWs spectral characteristics, potential detection prospects of these waves, and possibility to reconstruct (or limit) physical conditions much earlier than BBN [10–12].

There are several conditions, under which primordial turbulence can develop (turbulence can develop if there is a mechanism injecting kinetic energy, whatever a source of sufficiently large-scale motions exists) [9, 13–15]. Turbulence is a complex nonlinear process controlled by multiple physical parameters, would it be produced in a laboratory or in the early universe. These parameters depend on each other, and the degeneracy between these parameters further complicates the appropriate modeling [16–18]. After inflation, when thermalized plasma has been formed, the fluid with ultra relativistic equation of state $p = \rho/3$ with p - pressure, and ρ energy density) is characterized by two numbers, the hydrodynamic Reynolds number, $\mathrm{Re} = vL/\nu$ and the magnetic Reynolds number, $\mathrm{Rm} = vL/\eta$ where v is the characteristic velocity, L is the characteristic length scale, ν is the kinematic viscosity, and $\eta = \sigma^{-1}$ is the magnetic diffusivity (in natural units), with σ denoting the electrical conductivity. Their ratio defines the magnetic Prandtl number $\mathrm{Pm} = \mathrm{Rm}/\mathrm{Re} = \nu/\eta$.

During the radiation-dominated era, the primordial plasma was an excellent electrical conductor, resulting in an extremely small magnetic diffusivity. Consequently, the magnetic Reynolds number was many orders of magnitude larger than unity. The hydrodynamic Reynolds number was likewise very large, provided sufficient large-scale velocity perturbations were present, such as those generated during cosmological phase transitions or by chiral plasma instabilities. At the electroweak epoch ($T \sim 100\,\mathrm{GeV}$), viscosity is dominated by weakly interacting particles (mainly neutrinos before decoupling), and typical estimates are $\mathrm{Re} \sim 10^{10}$–10^{13}, $\mathrm{Rm} \sim 10^{16}$–10^{20} yielding a magnetic Prandtl number $\mathrm{Pm} \sim 10^{6}$–10^{8}. Thus, $\mathrm{Rm} \gg \mathrm{Re} \gg 1$. A similar hierarchy holds at the QCD phase transition ($T \sim 150\,\mathrm{MeV}$), where estimates give $\mathrm{Re} \sim 10^{7}$–10^{10}, $\mathrm{Rm} \gtrsim 10^{17}$, again implying $\mathrm{Pm} \gg 1$ [19–23].

It should be highlighted that neither the Reynolds number nor the magnetic Reynolds number is a fundamental parameter. Both depend on the characteristic velocity and length scale of the flow. The values quoted above correspond to motions on the energy-containing (stirring) scale, typically assumed to be a fraction of the Hubble radius at the corresponding epoch. While the precise numerical values therefore depend on the plasma properties, it is possible safely assume that both Reynolds numbers are overwhelmingly larger than unity throughout the radiation era. nonlinear advection dominates over viscous dissipation, leading magnetic fields remain effectively frozen into the plasma, and ideal magnetohydrodynamic (MHD) turbulence laws apply, including inverse transfer and inverse cascade processes for helical fields. Obviously, direct numerical simulations (DNS) cannot reproduce the realistic physical parameters and instead rely on scaling laws to extrapolate from accessible Reynolds numbers [24–27].

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